

The Math of Guitar

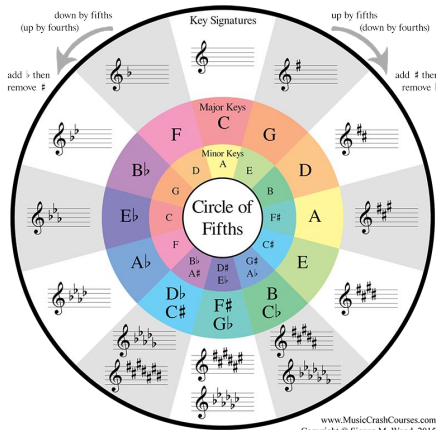
Ryan Y. Batubara

Icebreaker

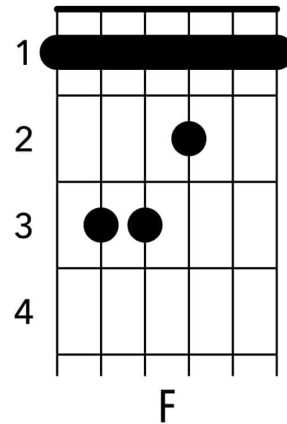
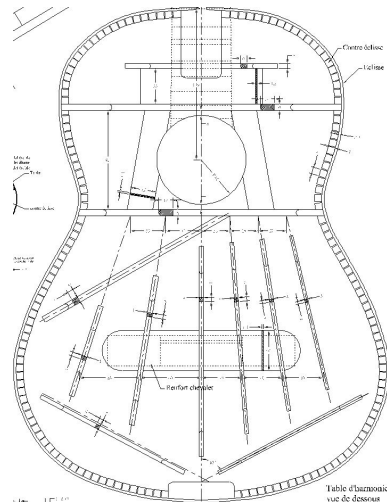
Which can produce a note of higher frequency, this guitar or piano?



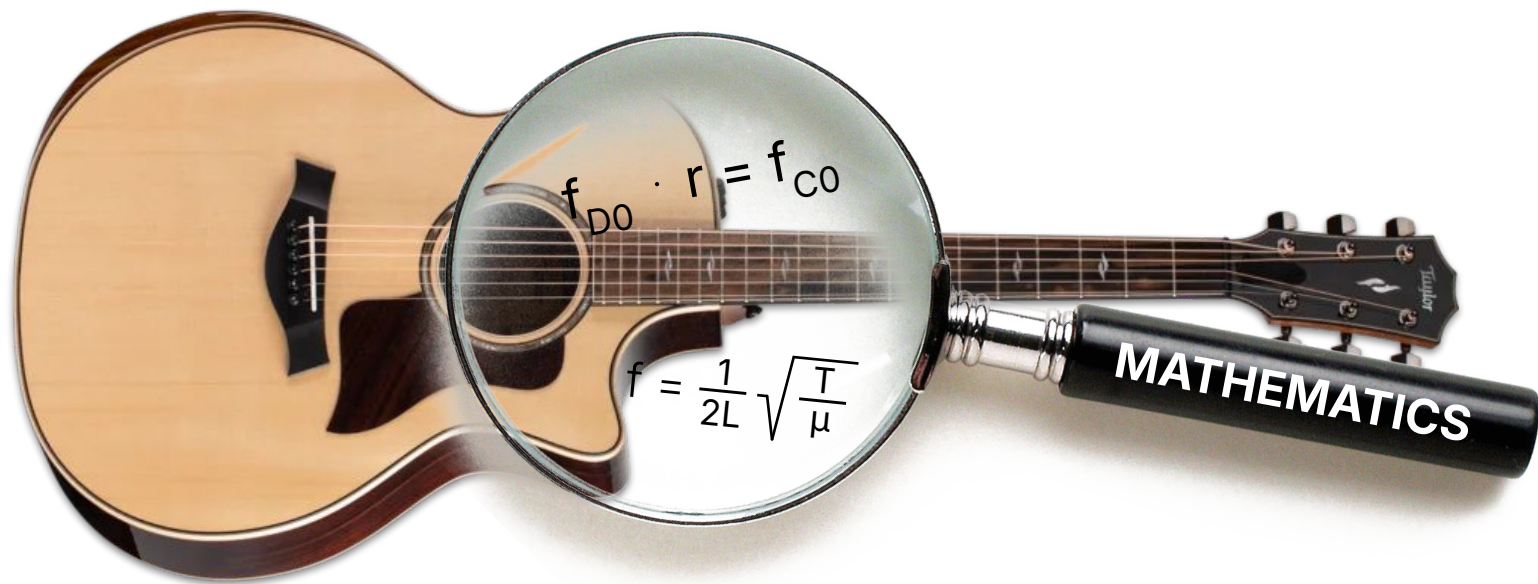
What this talk is NOT about



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What this talk is about



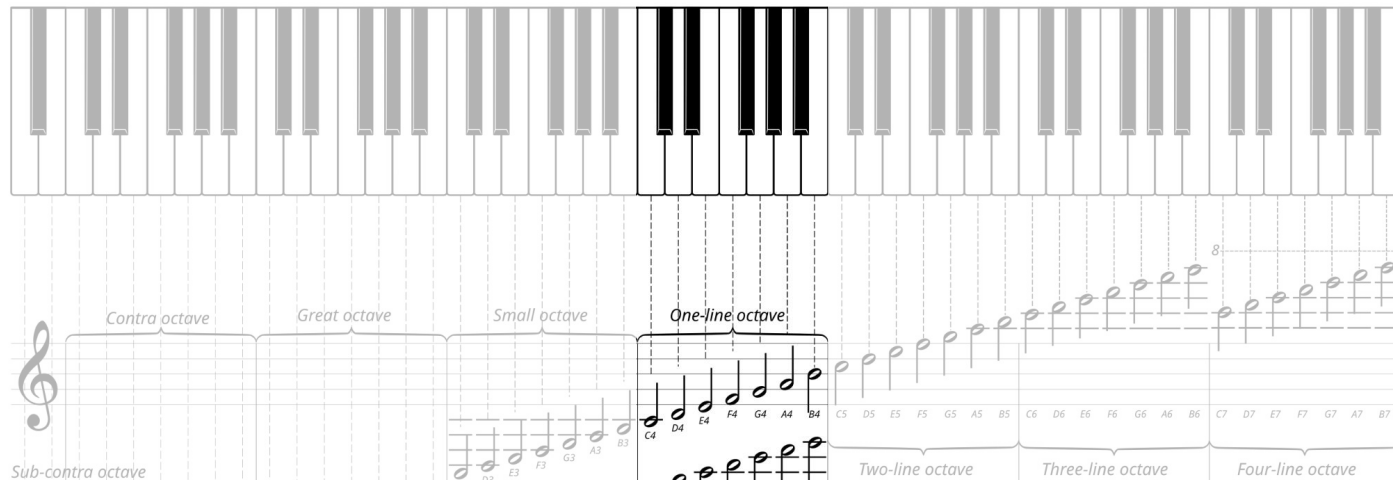
1. Models of Sound

The 12-tone system

There are 12 notes:

C C# D D# E F F# G G# A A# B

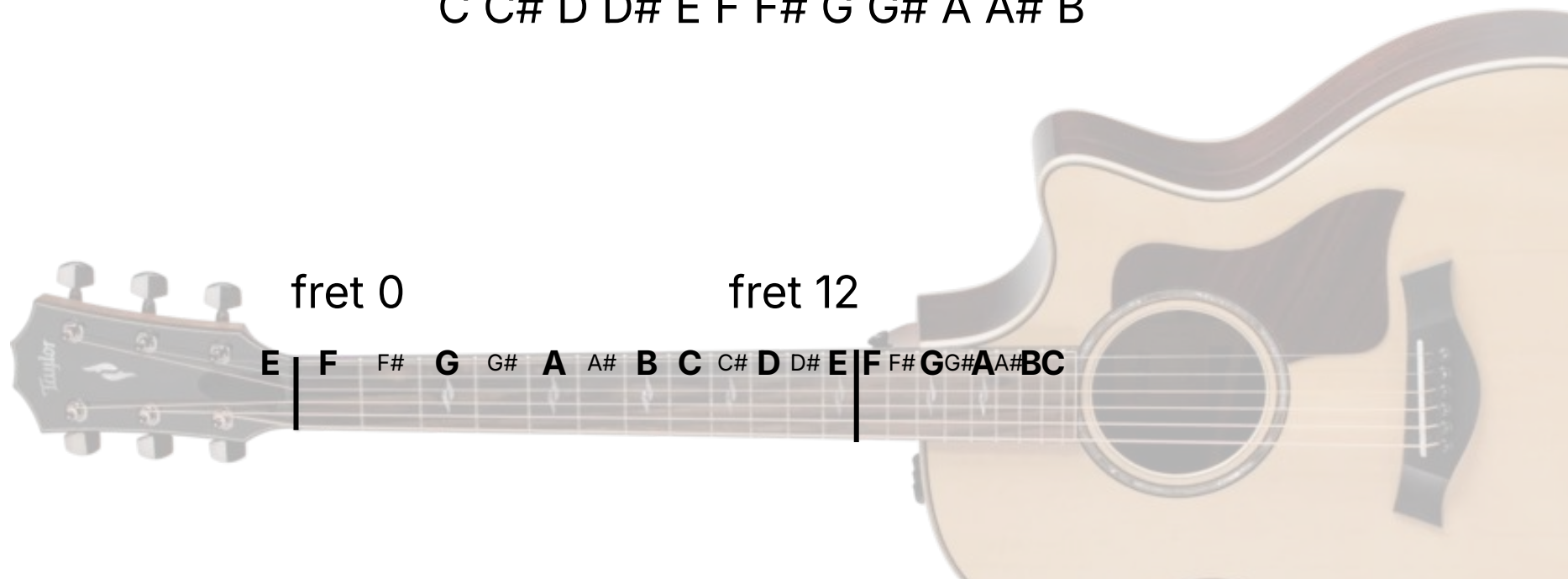
Each note is given an *octave*. Here is C4 up to B4:



The 12-tone system

There are 12 notes:

C C# D D# E F F# G G# A A# B

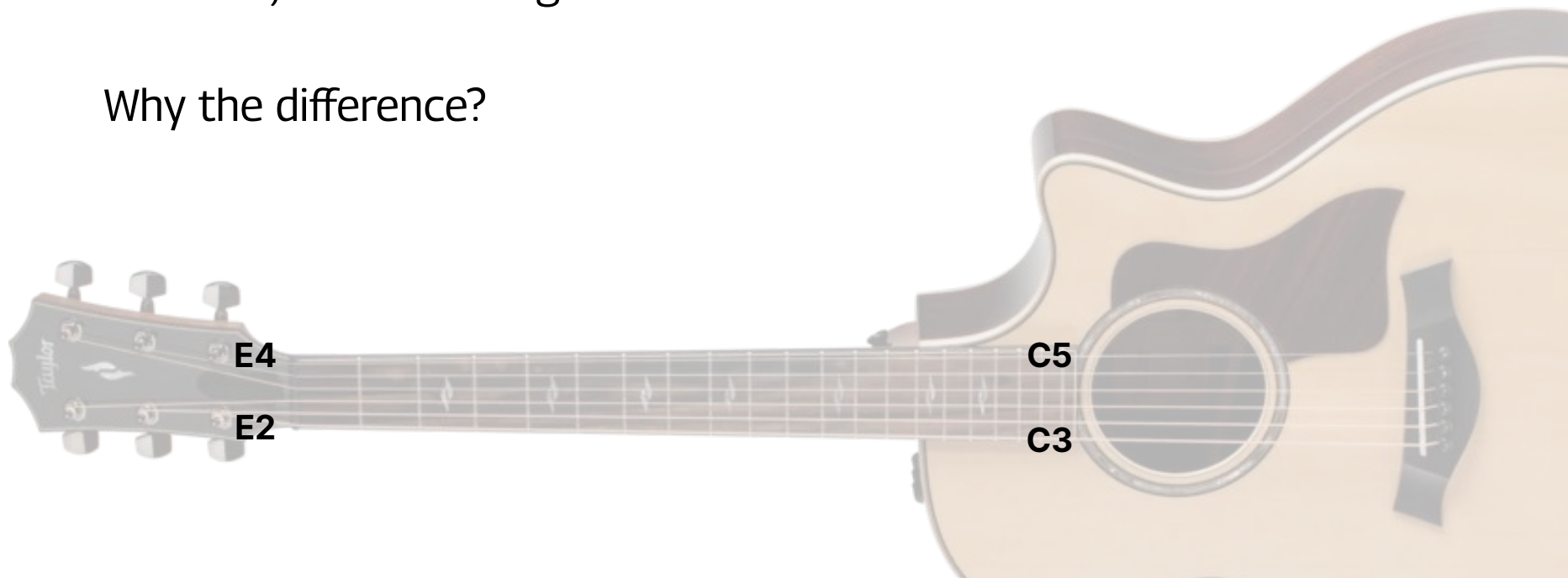


The 6 strings

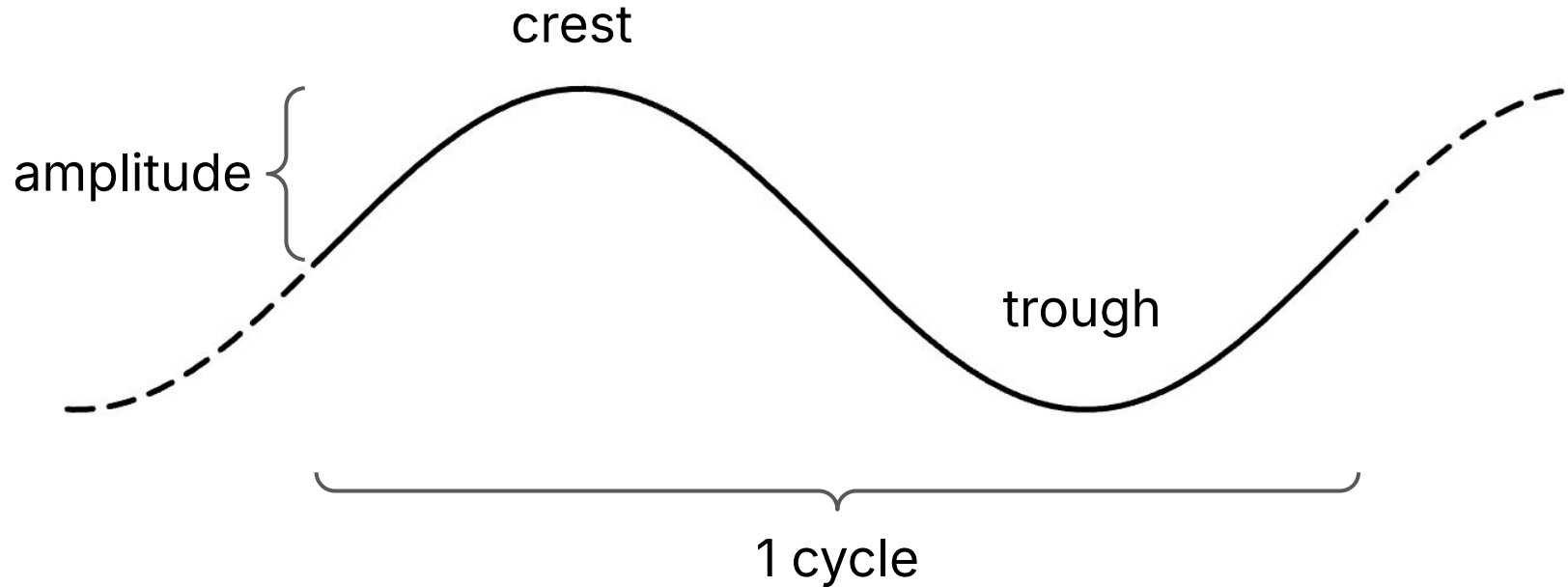
The *first*, top string has notes E4 ... C5.

The *sixth*, bottom string has notes E2 ... C3.

Why the difference?

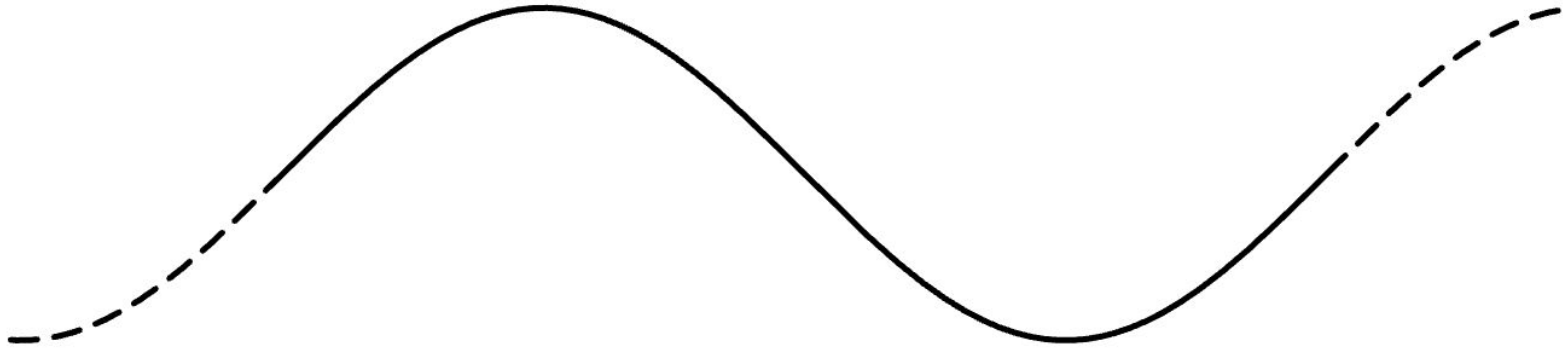


The transverse wave



$$v = f\lambda$$

speed = frequency $\times$ wavelength
m/s cycles/s m/cycle



wavelength = width of 1 cycle

The string frequency formula

The diagram shows the string frequency formula $f = \frac{1}{2L} \sqrt{\frac{T}{\mu}}$ with three labels and arrows pointing to its components:

- frequency** (cycles/s or Hz) points to f .
- length** (m) points to $2L$.
- tension** (kgm/s² or N) points to T .
- mass / length** (kg/m) points to μ .

$$f = \frac{1}{2L} \sqrt{\frac{T}{\mu}}$$

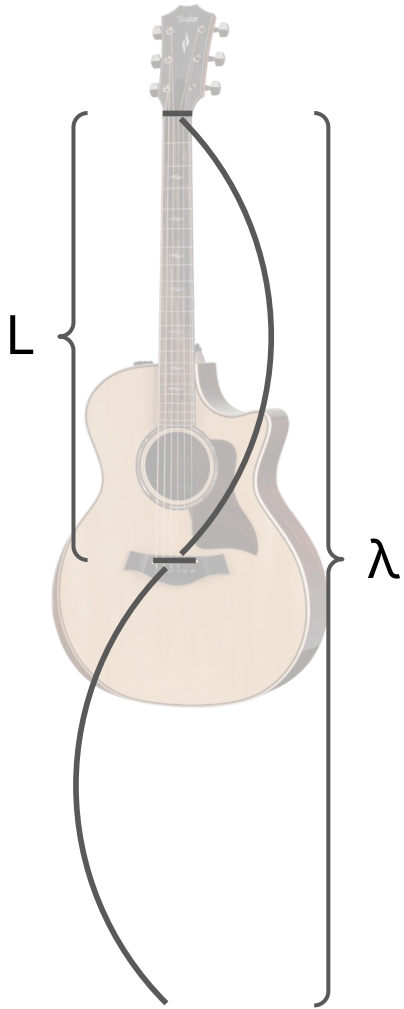
Why this formula?

$$f = \underbrace{\frac{1}{2L}} \underbrace{\sqrt{\frac{T}{\mu}}}$$

$v = f\lambda$
(rearranged)

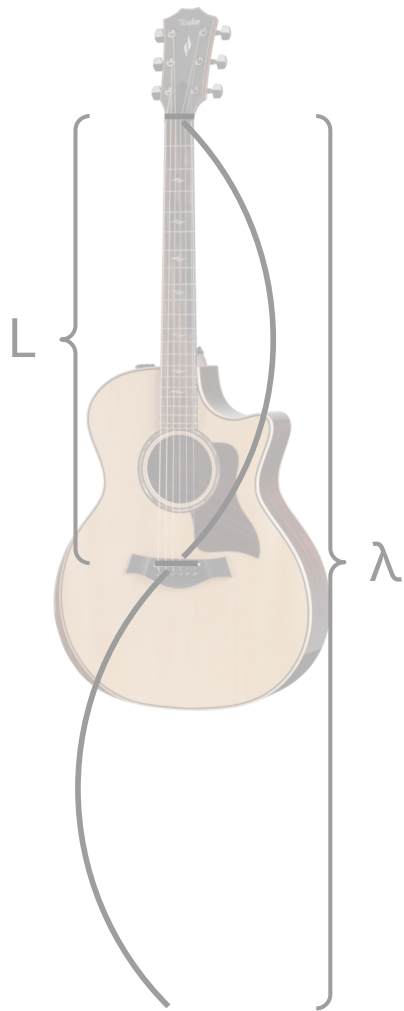

$$f = \frac{1}{\lambda} \cdot v$$

Why this formula?



$$f = \underbrace{\frac{1}{2L}} \underbrace{\sqrt{\frac{T}{\mu}}}$$
$$f = \frac{1}{\lambda} \cdot v$$

Why this formula?



$$f = \underbrace{\frac{1}{2L}}_{\frac{1}{\lambda}} \underbrace{\sqrt{\frac{T}{\mu}}}_v$$
$$f = \frac{1}{\lambda} \cdot v$$

$$v \propto \frac{T}{\mu}$$

$$\text{ms}^{-1} = \frac{(\text{kgms}^{-2})^a}{(\text{kgm}^{-1})^b}$$

$$\Rightarrow a = 1/2, \\ b = -1/2$$

The formal derivation

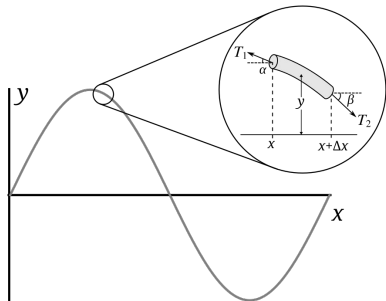
Let Δx be the **length** of a piece of string, m its **mass**, and μ its **linear density**. If angles α and β are small, then the horizontal components of **tension** on either side can both be approximated by a constant T , for which the net horizontal force is zero. Accordingly, using the **small angle approximation**, the horizontal tensions acting on both sides of the string segment are given by

$$T_{1x} = T_1 \cos(\alpha) \approx T.$$

$$T_{2x} = T_2 \cos(\beta) \approx T.$$

From Newton's second law for the vertical component, the mass (which is the product of its linear density and length) of this piece times its acceleration, a , will be equal to the net force on the piece:

$$\Sigma F_y = T_{1y} - T_{2y} = -T_2 \sin(\beta) + T_1 \sin(\alpha) = \Delta m a \approx \mu \Delta x \frac{\partial^2 y}{\partial t^2}.$$



Dividing this expression by T and substituting the first and second equations obtains (we can choose either the first or the second equation for T , so we conveniently choose each one with the matching angle β and α)

$$-\frac{T_2 \sin(\beta)}{T_2 \cos(\beta)} + \frac{T_1 \sin(\alpha)}{T_1 \cos(\alpha)} = -\tan(\beta) + \tan(\alpha) = \frac{\mu \Delta x}{T} \frac{\partial^2 y}{\partial t^2}.$$

According to the small-angle approximation, the tangents of the angles at the ends of the string piece are equal to the slopes at the ends, with an additional minus sign due to the definition of α and β . Using this fact and rearranging provides

$$\frac{1}{\Delta x} \left(\left. \frac{\partial y}{\partial x} \right|^{x+\Delta x} - \left. \frac{\partial y}{\partial x} \right|^x \right) = \frac{\mu}{T} \frac{\partial^2 y}{\partial t^2}.$$

In the limit that Δx approaches zero, the left hand side is the definition of the second derivative of y ,

$$\frac{\partial^2 y}{\partial x^2} = \frac{\mu}{T} \frac{\partial^2 y}{\partial t^2}.$$

this equation is known as the **wave equation**, and the coefficient of the second time derivative term is equal to $\frac{1}{v^2}$; thus

$$v = \sqrt{\frac{T}{\mu}},$$

Once the speed of propagation is known, the **frequency** of the **sound** produced by the string can be calculated. The **speed** of propagation of a wave is equal to the **wavelength** λ divided by the **period** τ , or multiplied by the **frequency** f :

$$v = \frac{\lambda}{\tau} = \lambda f.$$

If the length of the string is L , the **fundamental harmonic** is the one produced by the vibration whose **nodes** are the two ends of the string, so L is half of the wavelength of the fundamental harmonic. Hence one obtains **Mersenne's laws**:

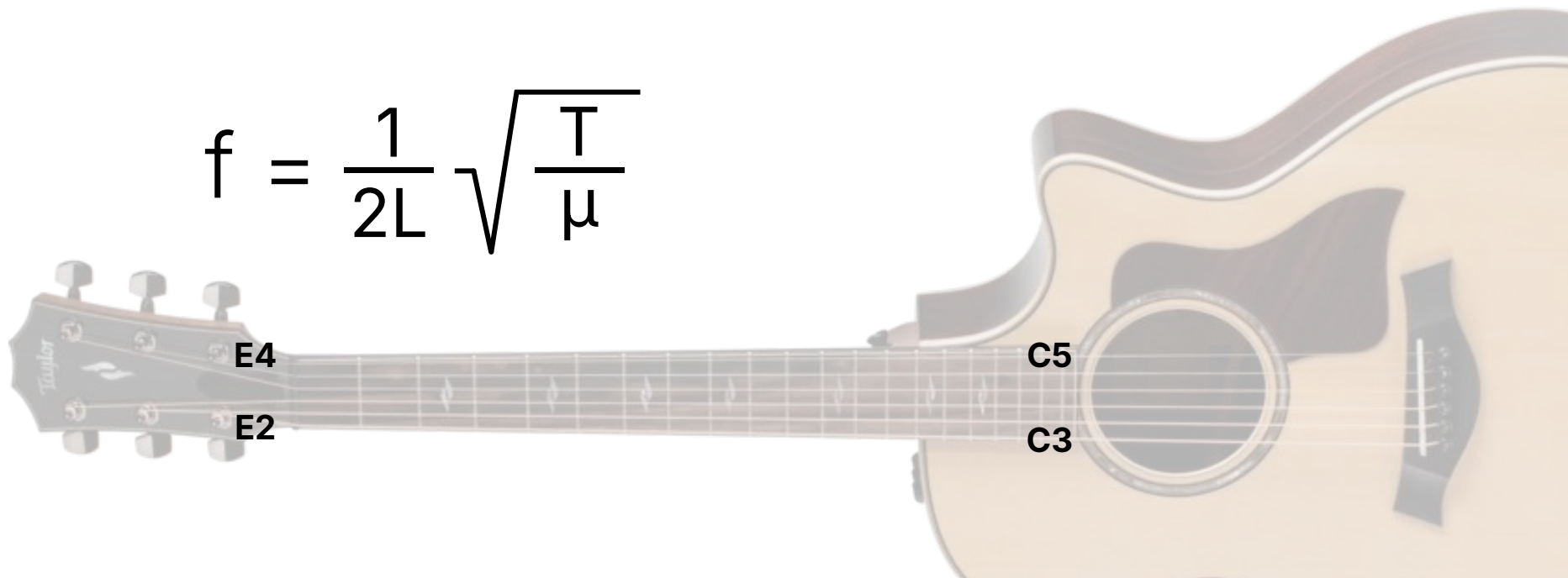
$$f = \frac{v}{2L} = \frac{1}{2L} \sqrt{\frac{T}{\mu}}$$

Back to the 6 strings

Why do different strings produce different notes?

A: Different tension and mass.

$$f = \frac{1}{2L} \sqrt{\frac{T}{\mu}}$$

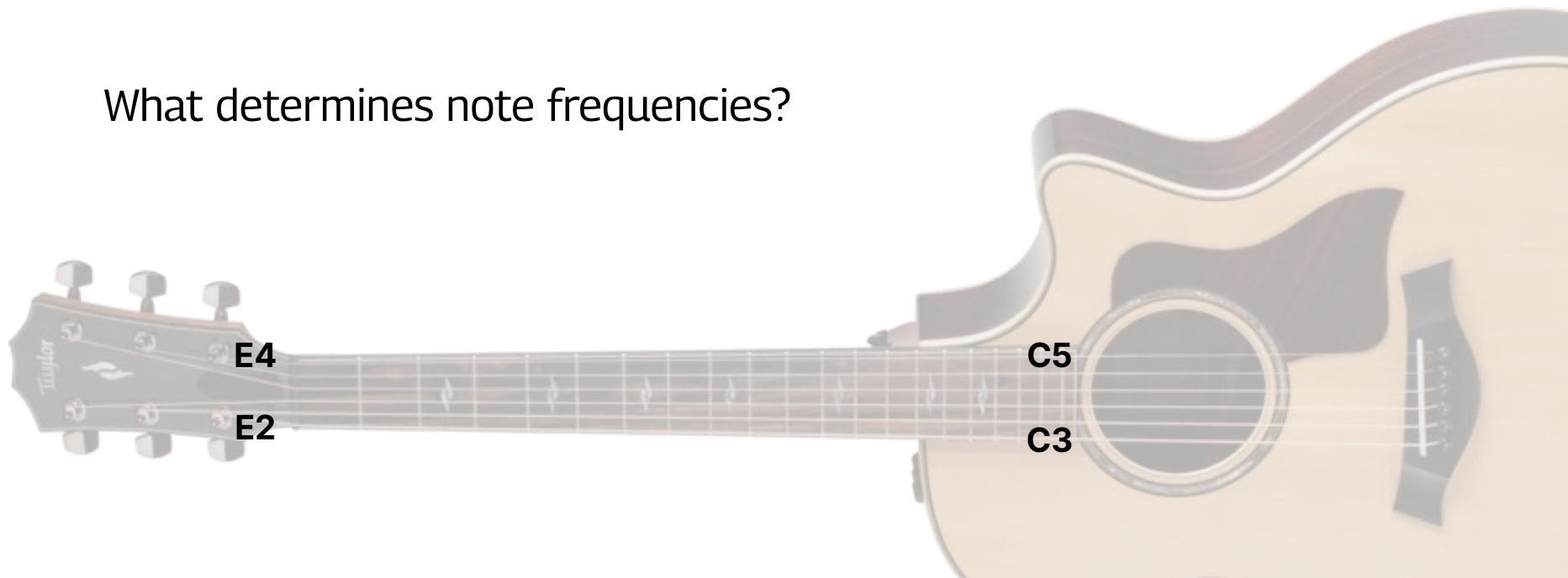


Back to the 6 strings

Why do different strings produce different notes?

A: Different tension and mass.

What determines note frequencies?

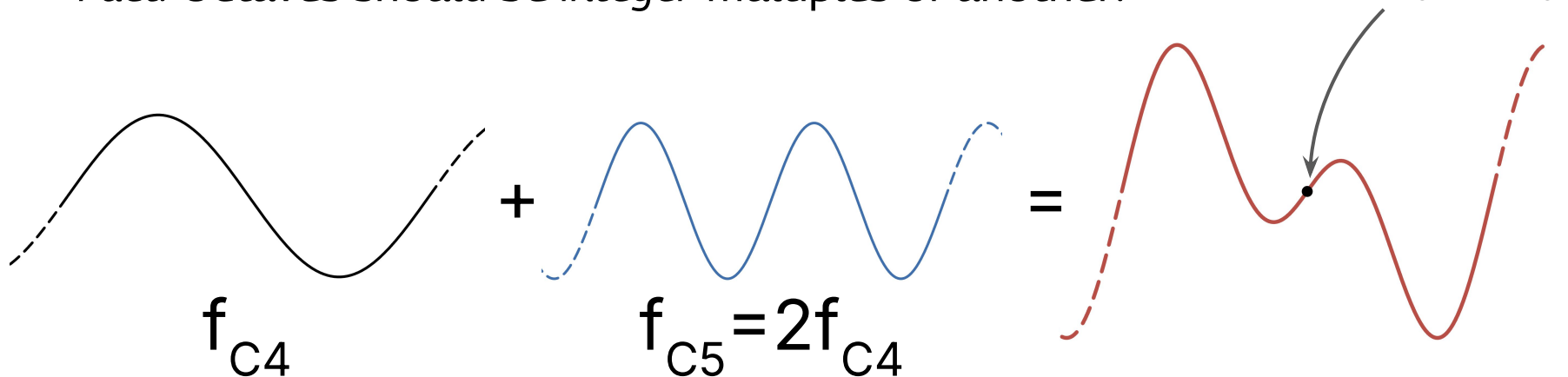


Back to notes

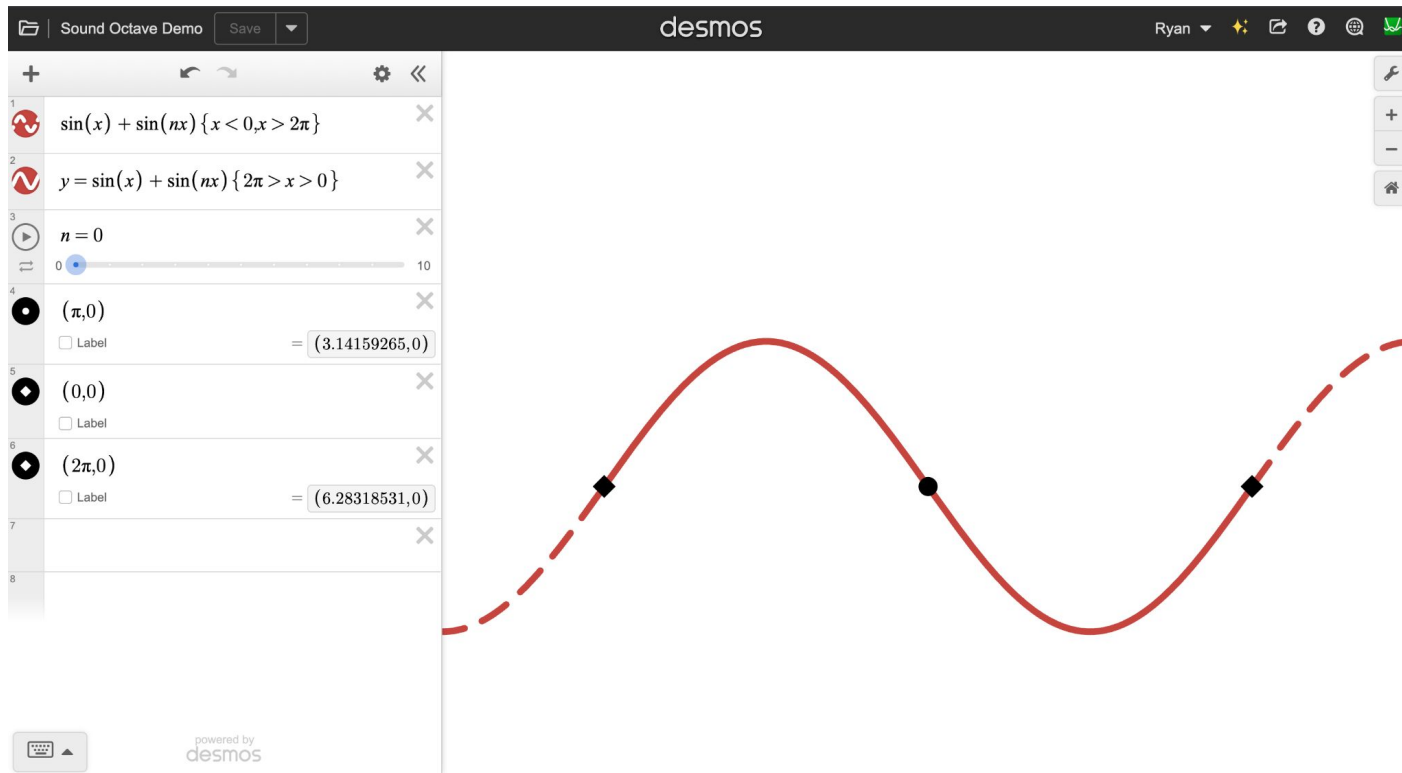
Each note and octave is just a certain frequency.

... B3 C4 C#4 ... B4 C5 C#5 ...
... f_{B3} f_{C4} $f_{C\#4}$... f_{B4} f_{C5} $f_{C\#5}$...

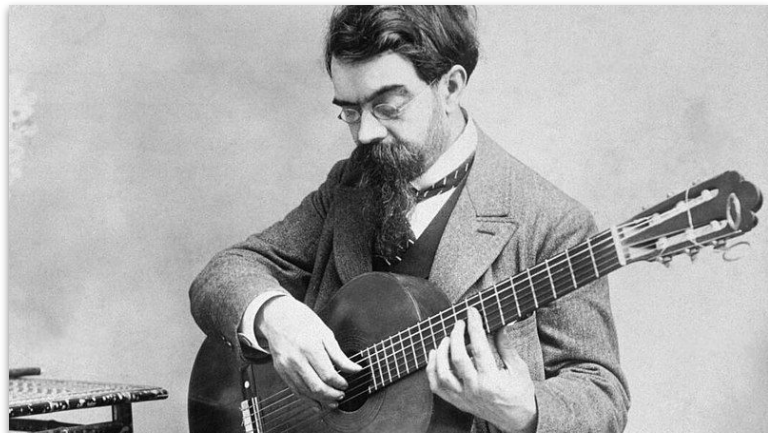
Fact: Octaves should be *integer* multiples of another:



Octaves demo



ex: octave intervals



"Tango"
(Francisco Tárrega)
DGDGBE

Musical notation for the piece "Tango" by Francisco Tárrega, showing three staves with notes and fingerings, and a corresponding fretboard diagram below.

The notation includes circled numbers 3, 2, and 5, indicating specific fingerings or positions. The fretboard diagram shows the fret numbers for each string (D, G, D, G, B, E) across two measures.

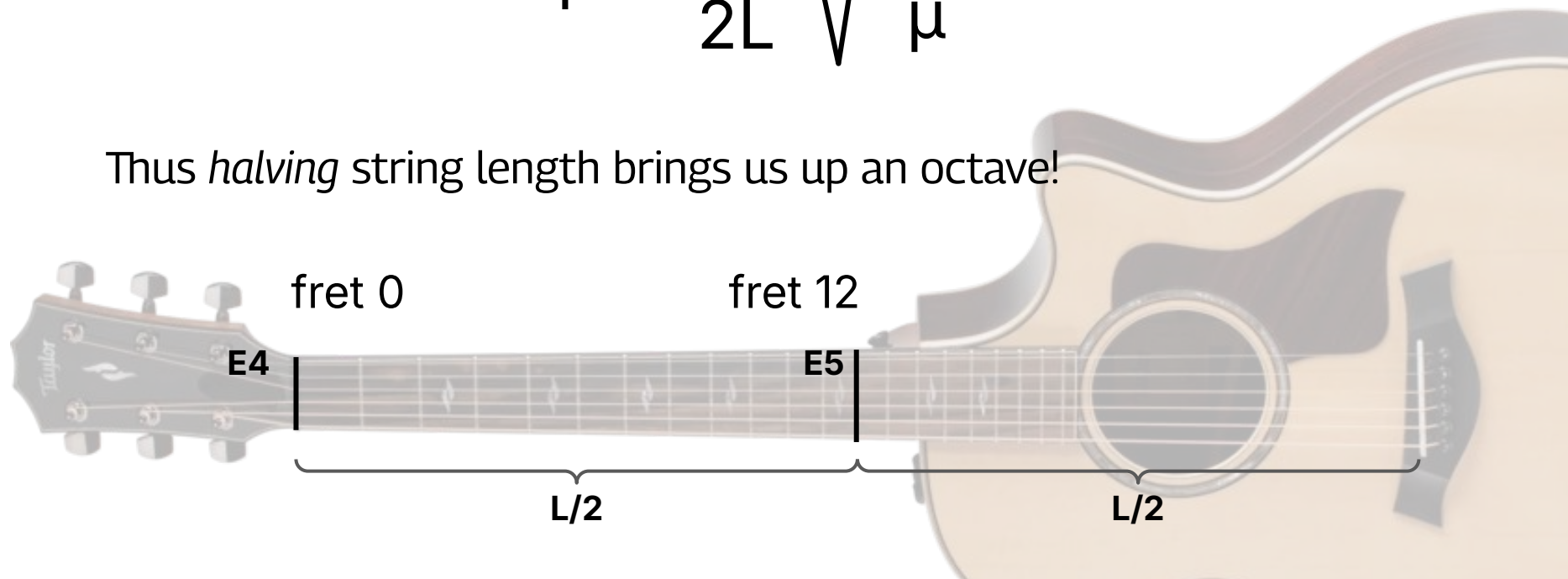
Measure 1	Measure 2
7	0
5	0
3	0
2	2

Octaves on the fretboard

Recall:

$$f = \frac{1}{2L} \sqrt{\frac{T}{\mu}}$$

Thus *halving* string length brings us up an octave!

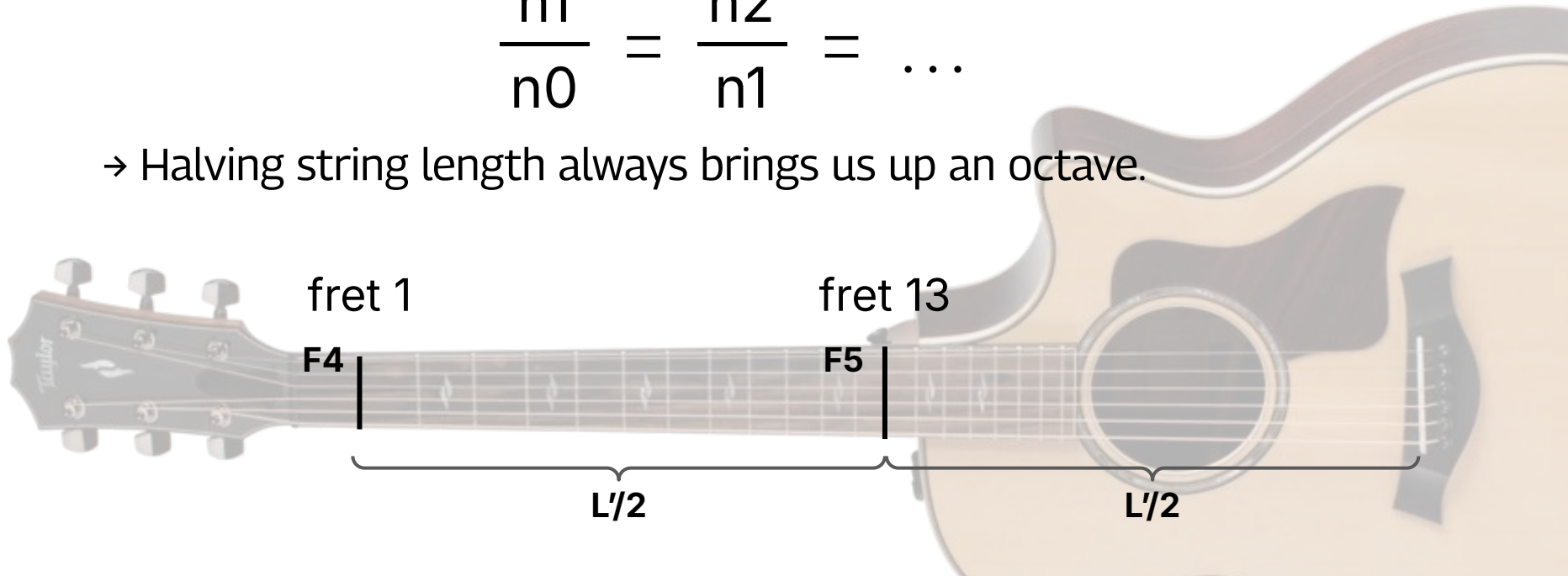


Other notes

What about other notes? For all notes n from C, C#, ..., we want

$$\frac{n_1}{n_0} = \frac{n_2}{n_1} = \dots$$

→ Halving string length always brings us up an octave.

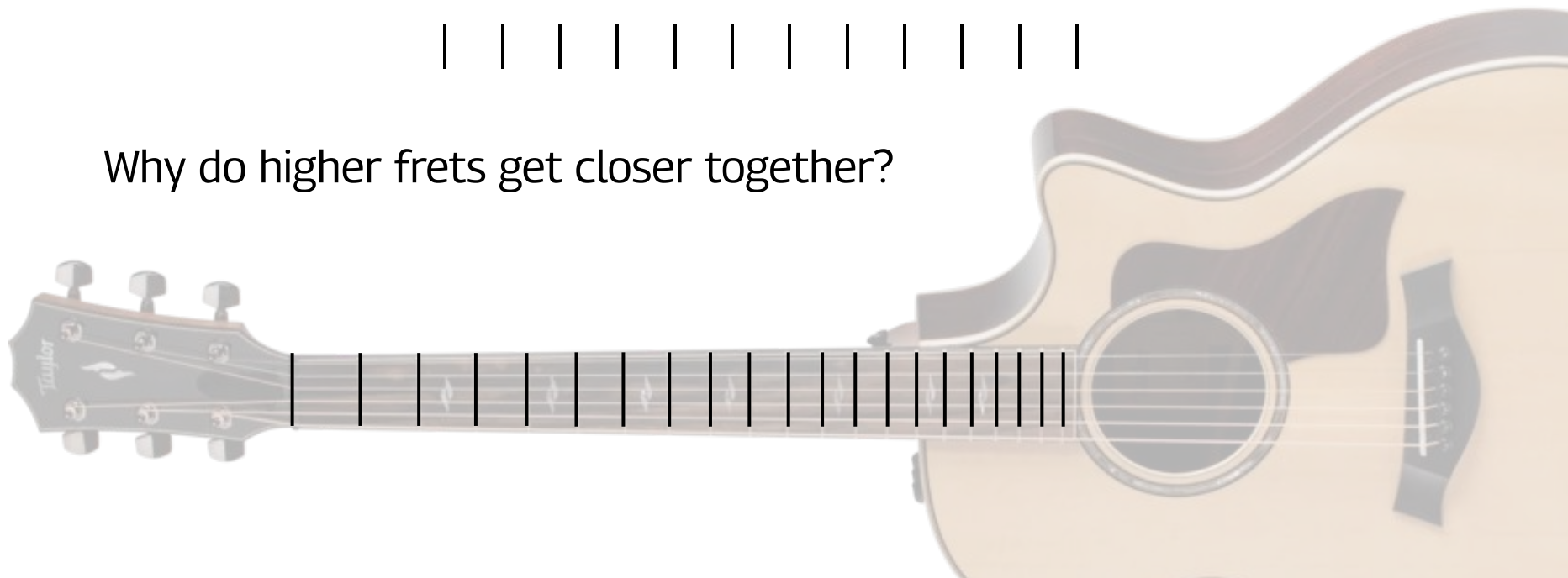


Fretboard design

So why aren't frets equidistant?



Why do higher frets get closer together?



Equal temperament

All notes should be *equally distant*. Suppose:

$$f_{D0} + c = f_{C0}$$

$$f_{D\#1} + c = f_{D0}$$

...

But we want

$$\frac{C1}{C0} = \frac{C0 + 12c}{C0} \neq \frac{C2}{C1} = \frac{C0 + 24c}{C0 + 12c}$$

Equal temperament

All notes should be *equally distant*. Suppose:

$$f_{D0} \cdot r = f_{C0}$$

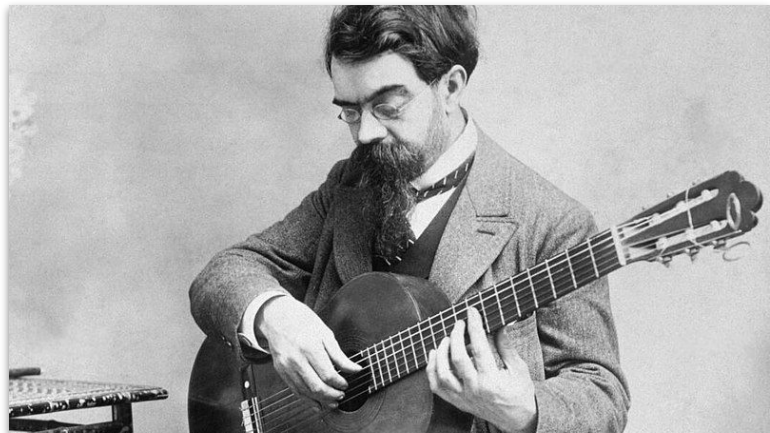
$$f_{D\#1} \cdot r = f_{D0}$$

...

Now we have

$$\frac{C1}{C0} = \frac{C0 \cdot r^{12}}{C0} = \frac{C2}{C1} = \frac{C0 \cdot r^{24}}{C0 \cdot r^{12}}$$

ex: other intervals



"Tango"
(Francisco Tárrega)
DGDGBE

Sheet music for the guitar piece "Tango" by Francisco Tárrega, showing two systems of music. The notation includes standard musical notation (treble and bass staves) and fretboard diagrams (fingerings and positions).

System 1:

- Staff 1 (Treble):** Contains musical notation with fingerings (1, 3, 1, 3, 1, 3) and a circled 1 above the first measure. A dashed line indicates a slur over the first two measures.
- Staff 2 (Bass):** Contains musical notation with fingerings (0, 0, 0, 0) and a circled 2 below the first measure.
- Fretboard Diagrams:** Shows positions 6-10, 8-11, 10-11, 10-11, (10-11), 10-11, 11-13, 11-13.

System 2:

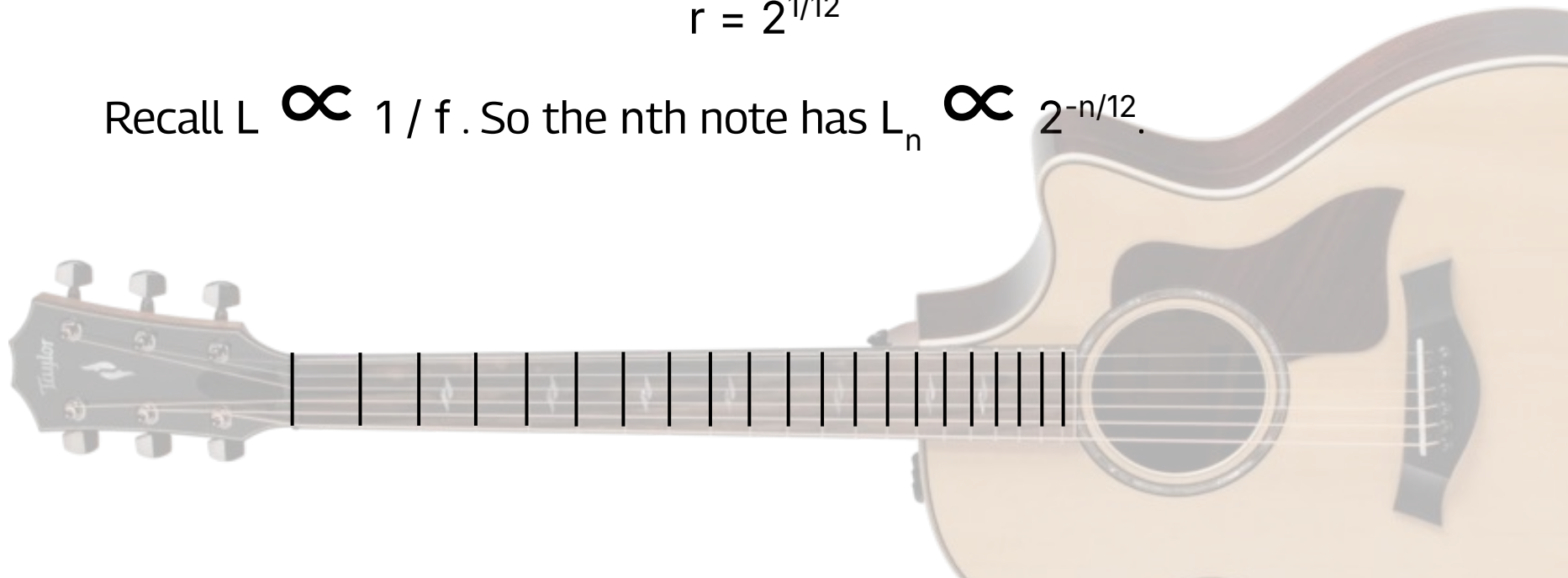
- Staff 1 (Treble):** Contains musical notation with fingerings (1, 1, 2, 1, 2, 1, 2, 1, 2) and a circled 1 above the first measure. A dashed line indicates a slur over the first two measures.
- Staff 2 (Bass):** Contains musical notation with fingerings (0, 0, 0, 0).
- Fretboard Diagrams:** Shows positions 13-11, 13-11, 10-11, 10-11, (10-11), 10-11, 11-13, 11-13.

Why uneven frets?

Since $f_{c_1} = 2 \cdot f_{c_0} = r^{12} \cdot f_{c_0}$ we have

$$r = 2^{1/12}$$

Recall $L \propto 1/f$. So the n th note has $L_n \propto 2^{-n/12}$.



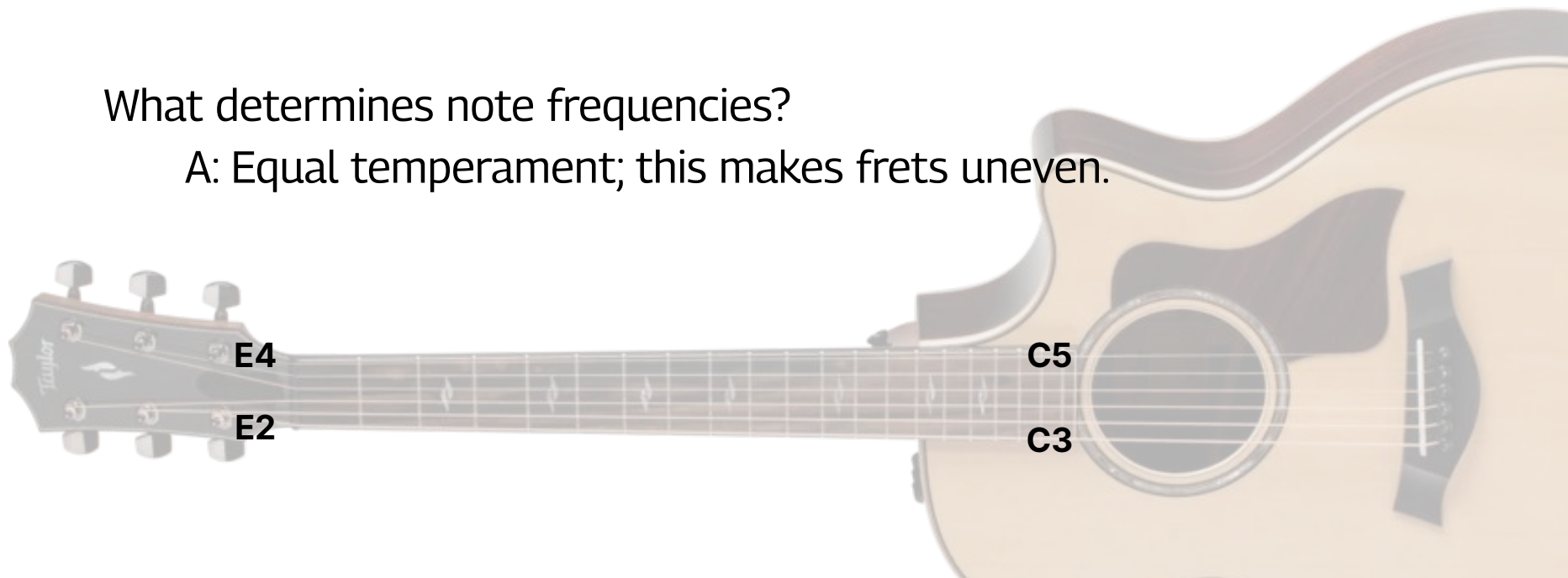
Back to the 6 strings

Why do different strings produce different notes?

A: Different tension and mass.

What determines note frequencies?

A: Equal temperament; this makes frets uneven.



Recall: icebreaker

Which can produce a note of higher frequency, this guitar or piano?

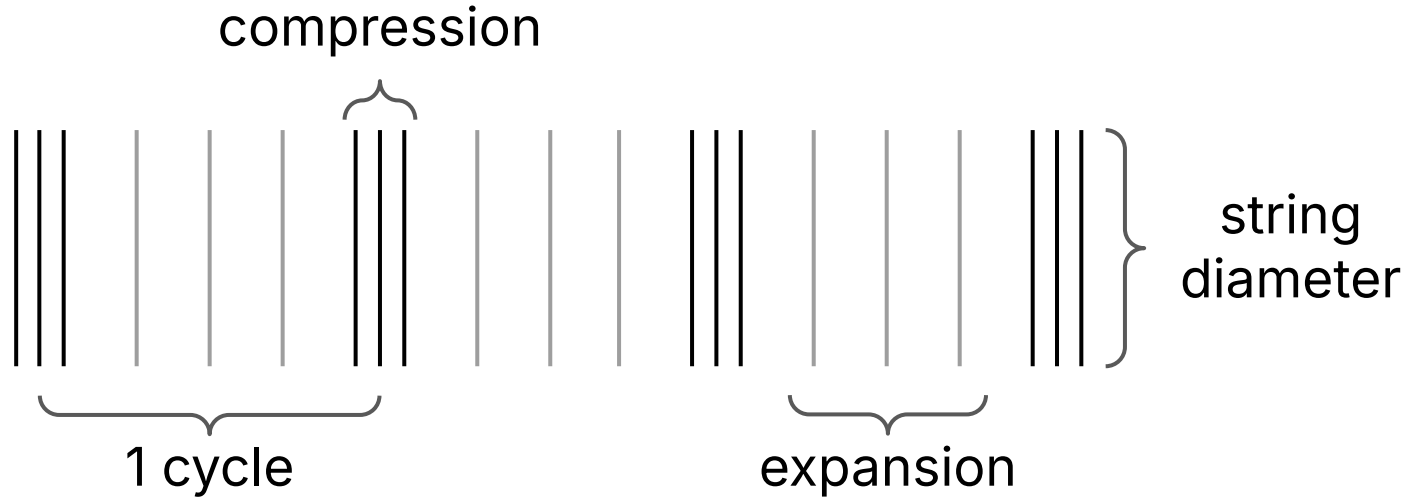
Guitar : highest string, highest fret is C5

Piano : highest note is C8



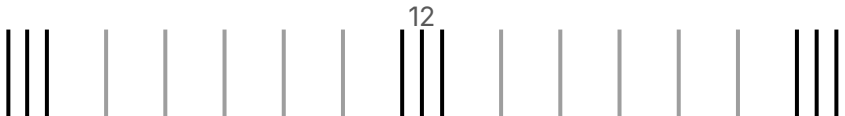
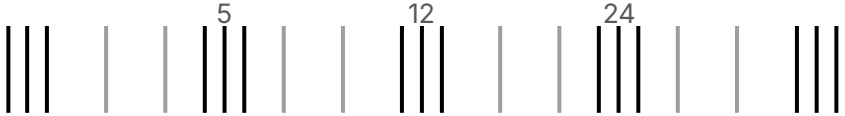
2. Harmonics

The longitudinal wave

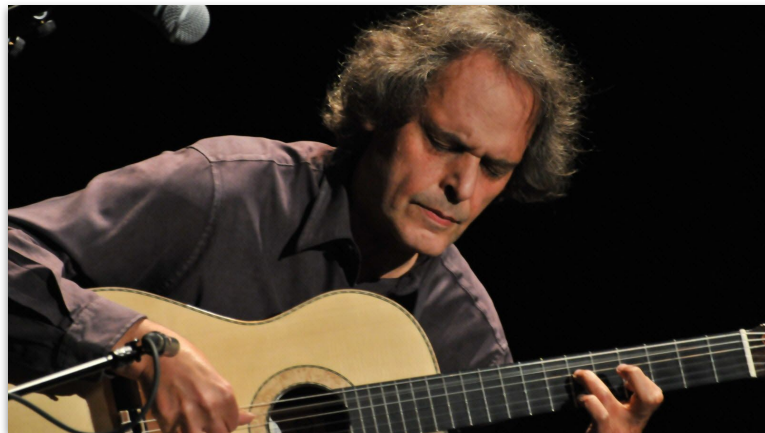


The harmonics

$$f = \frac{1}{2L} \sqrt{\frac{T}{\mu}}$$

1		f_1	fundamental
2		$f_2 = 2 f_1$	1 octave
3		$f_3 = 2 \cdot \frac{3}{2} f_1 \approx 2^3 \cdot 2^{4/12}$	1 oct. + 3rd
4		$f_4 = 2 \cdot 2 f_1$	2 octaves
5		$f_5 = 4 \cdot \frac{5}{4} f_1 \approx 2^5 \cdot 2^{7/12}$	2 oct. + 5th

ex: harmonics for reach



"Valse de Loges"
(Roland Dyens)
EADGBE

A musical score for guitar, showing a sequence of chords and a melodic line. The chords are labeled with Roman numerals: IV, III, III, VII, and 8va (XXIV). The notation includes fingerings (4, 5, 3, 6), accidentals (sharps), and dynamic markings (p, ppp). The piece is in EADGBE tuning. The duration is approximately 4'45.

④ — IV III ⑤ III ③ VII 8va (XXIV)

0 0 0 0 0

p, ma chiaro (pont.)

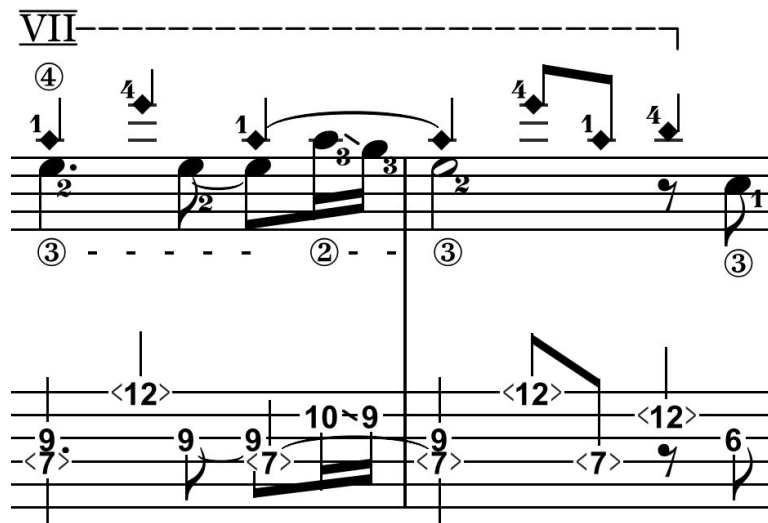
ppp (poetico)

durée ≈ 4'45

ex: harmonic melody



"Safe Haven"
(Sönke Meinen)
DADGBE



ex: harmonics for sustain



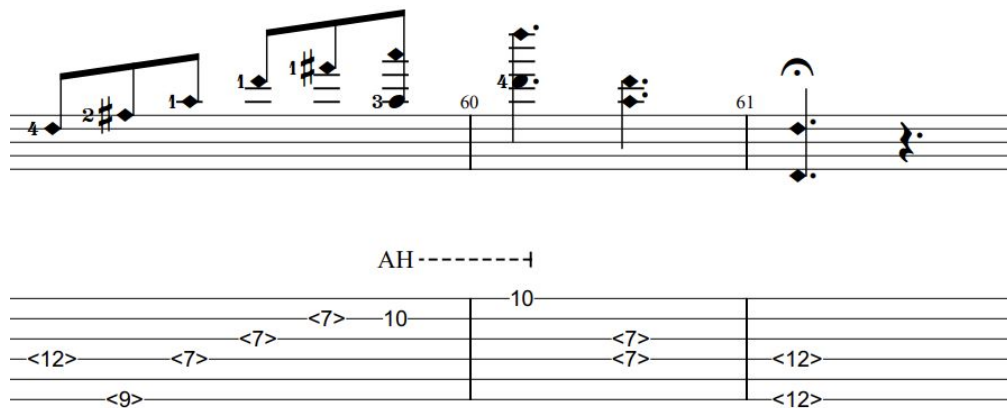
"Julia Florida"
(Agustín Barrios Mangoré)
DADGBE



ex: harmonics for scales



"Julia Florida"
(Agustín Barrios Mangoré)
DADGBE



ex: harmonics for volume



"Sunset on the Peak"
(Masaaki Kishibe 岸部真明)
DADF#AD

Open D Tuning
① = D ④ = D
② = A ⑤ = A
③ = F# ⑥ = D
Capo. 2 fret



ex: everything combined



"Bell's Harmonics"
(Alan Gogoll)
EAC#EBE

AH
0 <19> AH AH

0 0 0 <5> 0 <19> 0 <19>

0 0 <5> <5> <5>

Recall: icebreaker

Which can produce a note of higher frequency, this guitar or piano?

Guitar : highest string, fret, **& harmonic is C7**

Piano : highest note is C8



Closing question

Which can produce a note of **lower** frequency, this guitar or piano?

Guitar : ???

Piano : A0 (typical 88 key)

